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Heat pump control systems and methods

Abstract

A heat pump for a heating, ventilation, and air conditioning (HVAC) system includes a compressor system configured to direct a refrigerant flow along a refrigerant circuit of the heat pump. The compressor system includes a first compressor and a second compressor arranged in parallel with one another relative to a direction of the refrigerant flow through the compressor system, the first compressor includes a first volume index, and the second compressor includes a second volume index different from the first volume index. The heat pump also includes a controller communicatively coupled to the first compressor and the second compressor and configured to selectively operate the first compressor, the second compressor, or both based on an operating mode the heat pump.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority from and the benefit of U.S. Provisional Application No. 63/304,415, entitled “HEAT PUMP CONTROL SYSTEMS AND METHODS,” filed Jan. 28, 2022, which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND

(1) This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure, which are described below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

(2) A heating, ventilation, and/or air conditioning (HVAC) system may be used to thermally regulate an environment, such as a space within a building, home, or other structure. The HVAC system generally includes a vapor compression system having heat exchangers, such as a condenser and an evaporator, which transfer thermal energy between the HVAC system and the environment. Typically, a compressor is fluidly coupled to a refrigerant circuit of the vapor compression system and is configured to circulate a working fluid (e.g., refrigerant) between the condenser and the evaporator. In this way, the compressor facilitates heat exchange between the refrigerant, the condenser, and the evaporator. In some cases, refrigerant flow through the refrigerant circuit may be reversible, such that the condenser is operable as an evaporator (e.g., a heat absorber), and the evaporator is operable a condenser (e.g., a heat rejector). Accordingly, the HVAC system may operate as a heat pump system in multiple operating modes (e.g., a cooling mode, a heating mode) to provide both heating and cooling to the building with one refrigeration circuit. Unfortunately, conventional compressors may be ill-suited to selectively operate the refrigerant circuit in the cooling mode or the heating mode. Therefore, implementation of such compressors in heat pump systems may reduce an overall operational efficiency of the HVAC system.

SUMMARY

(3) A summary of certain embodiments disclosed herein is set forth below. It should be noted that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

(4) The present disclosure relates to a heat pump for a heating, ventilation, and air conditioning (HVAC) system including a compressor system configured to direct a refrigerant flow along a refrigerant circuit of the heat pump. The compressor system includes a first compressor and a second compressor arranged in parallel with one another relative to a direction of the refrigerant flow through the compressor system, the first compressor includes a first volume index, and the second compressor includes a second volume index different from the first volume index. The heat pump also includes a controller communicatively coupled to the first compressor and the second compressor and configured to selectively operate the first compressor, the second compressor, or both based on an operating mode the heat pump.

(5) The present disclosure also relates to a heat pump for a heating, ventilation, and air conditioning (HVAC) system including a first compressor configured to direct refrigerant flow along a refrigerant circuit of the heat pump, where the first compressor has a first built-in volume ratio, and a second compressor configured to direct refrigerant flow along the refrigerant circuit of the heat pump, where the second compressor has a second built-in volume ratio. The first compressor and the second compressor are arranged in parallel with one another relative to a direction of refrigerant flow through the refrigerant circuit, and the second built-in volume ratio is greater than the first built-in volume ratio. The heat pump also includes a controller communicatively coupled to the first compressor and the second compressor and configured to selectively operate the first compressor, the second compressor, or both based on an operating mode the heat pump,

(6) The present disclosure further relates to a heat pump for a heating, ventilation, and air conditioning (HVAC) system including a refrigerant circuit and a compressor system disposed along the refrigerant circuit and configured to direct a refrigerant along the refrigerant circuit, where the compressor system includes a first compressor and a second compressor arranged in

parallel with one another relative to a flow direction of the refrigerant through the compressor system, the first compressor includes a first volume index, and the second compressor includes a second volume index greater than the first volume index. The heat pump further includes a first heat exchanger disposed along the refrigerant circuit and configured to place the refrigerant in a heat exchange relationship with a supply air flow, a second heat exchanger disposed along the refrigerant circuit and configured to place the refrigerant in a heat exchange relationship with an ambient air flow, and a reversing valve configured to direct the refrigerant from the compressor system to the second heat exchanger in a cooling mode of the heat pump and configured to direct the refrigerant from the compressor system to the first heat exchanger in a heating mode of the heat pump. The heat pump also includes a controller communicatively coupled to the first compressor and the second compressor and configured to operate the first compressor and suspend operation of the second compressor in the cooling mode.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings in which:
- (2) FIG. 1 is a perspective view of an embodiment of a building incorporating a heating, ventilation, and air conditioning (HVAC) system in a commercial setting, in accordance with an aspect of the present disclosure;
- (3) FIG. 2 is a perspective view of an embodiment of a packaged HVAC unit, in accordance with an aspect of the present disclosure;
- (4) FIG. 3 is a perspective view of an embodiment of a split, residential HVAC system, in accordance with an aspect of the present disclosure;
- (5) FIG. 4 is a schematic diagram of an embodiment of a vapor compression system used in an HVAC system, in accordance with an aspect of the present disclosure;
- (6) FIG. 5 is a schematic diagram of an embodiment of a portion of an HVAC system that includes a heat pump system, illustrating the heat pump system configured for operation in a cooling mode, in accordance with an aspect of the present disclosure;
- (7) FIG. 6 is a schematic diagram of an embodiment of a portion of an HVAC system that includes a heat pump system, illustrating the heat pump system configured for operation in a heating mode, in accordance with an aspect of the present disclosure;
- (8) FIG. 7 is a flow diagram of an embodiment of a process for operating a heat pump system, in accordance with an aspect of the present disclosure;
- (9) FIG. 8 is a flow diagram of an embodiment of a process for operating a heat pump system, in accordance with an aspect of the present disclosure;
- (10) FIG. 9 is a flow diagram of an embodiment of a process for operating a heat pump system, in accordance with an aspect of the present disclosure;
- (11) FIG. 10 is a flow diagram of an embodiment of a process for operating a heat pump system, in accordance with an aspect of the present disclosure;
- (12) FIG. 11 is a flow diagram of an embodiment of a process for operating a heat pump system, in accordance with an aspect of the present disclosure;
- (13) FIG. 12 is a flow diagram of an embodiment of a process for operating a heat pump system, in accordance with an aspect of the present disclosure;
- (14) FIG. 13 is a flow diagram of an embodiment of a process for operating a heat pump system, in accordance with an aspect of the present disclosure;
- (15) FIG. 14 is a schematic diagram of an embodiment of a portion of an HVAC system that includes a split heat pump system, illustrating the split heat pump system configured for operation

in a heating mode, in accordance with an aspect of the present disclosure;

(16) FIG. 15 is a schematic diagram of an embodiment of a portion of an HVAC system that includes a split heat pump system, illustrating the split heat pump system configured for operation in a heating mode, in accordance with an aspect of the present disclosure; and

(17) FIG. 16 is a schematic diagram of an embodiment of a portion of an HVAC system that includes a dual heat pump system, in accordance with an aspect of the present disclosure.

DETAILED DESCRIPTION

(18) One or more specific embodiments of the present disclosure will be described below. These described embodiments are examples of the presently disclosed techniques. Additionally, in an effort to provide a concise description of these embodiments, all features of an actual implementation may not be described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(19) When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to “one embodiment” or “an embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

(20) As used herein, the terms “approximately,” “generally,” and “substantially,” and so forth, are intended to convey that the property value being described may be within a relatively small range of the property value, as those of ordinary skill would understand. For example, when a property value is described as being “approximately” equal to (or, for example, “substantially similar” to) a given value, this is intended to mean that the property value may be within $\pm 5\%$, within $\pm 4\%$, within $\pm 3\%$, within $\pm 2\%$, within $\pm 1\%$, or even closer, of the given value. Similarly, when a given feature is described as being “substantially parallel” to another feature, “generally perpendicular” to another feature, and so forth, this is intended to mean that the given feature is within $\pm 5\%$, within $\pm 4\%$, within $\pm 3\%$, within $\pm 2\%$, within $\pm 1\%$, or even closer, to having the described nature, such as being parallel to another feature, being perpendicular to another feature, and so forth. Further, it should be understood that mathematical terms, such as “planar,” “slope,” “perpendicular,” “parallel,” and so forth are intended to encompass features of surfaces or elements as understood to one of ordinary skill in the relevant art, and should not be rigidly interpreted as might be understood in the mathematical arts. For example, a “planar” surface is intended to encompass a surface that is machined, molded, or otherwise formed to be substantially flat or smooth (within related tolerances) using techniques and tools available to one of ordinary skill in the art. Similarly, a surface having a “slope” is intended to encompass a surface that is machined, molded, or otherwise formed to be oriented at an angle (e.g., incline) with respect to a point of reference using techniques and tools available to one of ordinary skill in the art.

(21) As briefly discussed above, a heating, ventilation, and air conditioning (HVAC) system may be used to thermally regulate a space within a building, home, or other suitable structure. For example, the HVAC system may include a vapor compression system that operates to transfer thermal energy between a working fluid, such as a refrigerant, and a fluid to be conditioned, such as air. The vapor compression system includes heat exchangers, such as a condenser and an evaporator, which are fluidly coupled to one another via one or more conduits of a working fluid loop or circuit. A compressor may be used to circulate the working fluid through the conduits and other components

of the working fluid circuit (e.g., an expansion device) and, thus, enable the transfer of thermal energy between components of the working fluid circuit (e.g., between the condenser and the evaporator) and one or more thermal loads (e.g., an environmental air flow, a supply air flow). Additionally or alternatively, the HVAC system may include a heat pump (e.g., a heat pump system, a reverse-cycle heat pump) having a first heat exchanger (e.g., a heating and/or cooling coil, an indoor coil, the evaporator) positioned within or otherwise fluidly coupled to the space to be conditioned, a second heat exchanger (e.g., a heating and/or cooling coil, an outdoor coil, the condenser) positioned in or otherwise fluidly coupled to an ambient environment (e.g., the atmosphere), and a pump (e.g., the compressor) configured to circulate the working fluid (e.g., refrigerant) between the first and second heat exchangers to enable heat transfer between the thermal load and the ambient environment, for example. The heat pump system is operable to provide both cooling or heating to the space to be conditioned (e.g., a room, zone, or other region within a building) by adjusting a flow of the working fluid through the working fluid circuit.

(22) For example, during operation of the heat pump system in a cooling mode, the compressor may direct working fluid through the working fluid circuit and the first and second heat exchangers in a first flow direction. While receiving working fluid in the first flow direction, the first heat exchanger (which may be positioned within the space to be conditioned) may operate as an evaporator and, thus, enable working fluid flowing through the first heat exchanger to absorb thermal energy from an air flow directed to the space. Further, the second heat exchanger (which may be positioned in the ambient environment surrounding the heat pump system), may operate as a condenser to reject the heat absorbed by the working fluid flowing from the first heat exchanger (e.g., to an ambient air flow directed across the second heat exchanger). In this way, the heat pump system may facilitate cooling of the space or other thermal load serviced by (e.g., in thermal communication with) the first heat exchanger.

(23) Conversely, during operation in a heating mode, a reversing valve (e.g., a switch-over valve) enables the compressor to direct working fluid through the refrigerant circuit and the first and second heat exchangers in a second flow direction, opposite the first flow direction. While receiving working fluid in the second flow direction, the first heat exchanger may operate as a condenser instead of an evaporator, and the second heat exchanger may operate as an evaporator instead of a condenser. As such, the first heat exchanger may receive (e.g., from the second heat exchanger) a flow of heated working fluid to reject heat to thermal load serviced by the first heat exchanger (e.g., an air flow directed to the space) and, thus, facilitate heating of the thermal load. In this way, the heat pump system may facilitate either heating or cooling of the thermal load based on the selected operational mode of the heat pump system (e.g., based on a flow direction of working fluid along the working fluid circuit).

(24) In many cases, pressure differentials or pressure ratios across various components (e.g., a compressor) or sections of the working fluid circuit (e.g., refrigerant circuit) may vary based on the mode (e.g., cooling operating mode, heating operating mode) in which the heat pump system is operated. As an example, pressure ratios across a compressor of the working fluid circuit may be relatively small while the heat pump system operates in the cooling mode and may be relatively large while the heat pump system operates in the heating mode. In particular, such pressure ratios may be indicative of a differential between an entering working fluid pressure at an inlet of the compressor and an exiting working fluid pressure at an outlet of the compressor. Typically, a volume index (e.g., a volume ratio, built-in volume ratio) of the compressor coupled to the working fluid circuit may be fixed (e.g., invariable), which may cause the compressor to be ill-suited or incapable of adjusting working fluid compression and working fluid circulation along the working fluid circuit in response to the varying pressure differentials that may be encountered between operation in the cooling and heating modes of the heat pump system. As a result, throughout a duration in which the heat pump system operates in either the cooling mode, the heating mode, or both, conventional compressors may limit an overall operational efficiency of the HVAC system.

(25) Accordingly, embodiments of the present disclosure relate to a heat pump system having a compressor system configured to adjust compression and/or circulation of working fluid (e.g., refrigerant) along the working fluid circuit based on one or more operational parameters, modes, and/or characteristics of the heat pump system to improve efficiency of the heat pump system. Such operational parameters may include an operational mode (e.g., cooling, heating) of the heat pump system, a load or demand of one or more thermal loads serviced by the heat pump system, and/or ambient (e.g., atmospheric) conditions surrounding the heat pump system. In any case, implementation of the disclosed heat pump system may improve the overall operational efficiency of the HVAC system during cooling and heating operations.

(26) For example, the compressor system disclosed herein may include a first compressor and a second compressor that are fluidly coupled to the working fluid circuit (e.g., in a parallel configuration). The first compressor (e.g., one or more compressors) may include operational characteristics (e.g., a volume index, built-in volume ratio, compression ratio, a capacity, a power output) that facilitate enhanced operation of the heat pump system in the cooling mode, while the second compressor (e.g., one or more compressors) may include operational characteristics that facilitate enhanced operation of the heat pump system in the heating mode. As an example, the first compressor configured to enable more efficient operation of the heat pump system in the cooling mode may have a first built-in volume ratio (BVR) or volume index (e.g., fixed volume index) that is lower than a second BVR or volume index of the second compressor configured to enable more efficient operation of the heat pump system in the heating mode.

(27) In some embodiments, respective characteristics of the first and second compressors may be selected for a particular heat pump system based on an expected duration of different ambient temperatures across a particular time period (e.g., a season, a year). In other words, respective characteristics of the first and second compressors may be selected based on a particular climate (e.g., warm, temperate, cold) or geographic region in which the heat pump is to be utilized. For example, in a warmer climate, the first compressor may have a larger displacement than the second compressor, in a temperate climate the first compressor and the second compressor may have similar displacements, and in a colder climate, the second compressor may have a larger displacement than the first compressor. A controller of the heat pump system may be configured to selectively operate the first compressor, the second compressor, both the first compressor and the second compressor, and/or one or more additional compressors of the compressor system based on the operational parameters of the heat pump system and the operational characteristics of the compressors in a manner that enhances the overall operational efficiency of the heat pump system.

(28) As an example, upon receiving a call (e.g., a control instruction) to operate the heat pump system in the cooling mode, the controller may activate the first compressor while retaining remaining compressors of the compressor system (e.g., the second compressor) in an idle (e.g., inactive) state. Conversely, upon receiving a call to operate the heat pump system in the heating mode, the controller may activate the second compressor while retaining remaining compressors of the compressor system (e.g., the first compressor) in the idle state. In this way, the controller may operate particular compressors of the compressor system that enable an improved relative operational efficiency of the heat pump system based on the current operating conditions experienced by the heat pump system (e.g., based on the current operating mode of the heat pump system).

(29) As discussed in detail below, the controller may selectively operate individual compressors or combinations of compressors included in the compressor system in accordance with the presently disclosed techniques. Moreover, it should be understood that one or more of the compressors included in the compressor system may be fixed speed compressors, multi-stage (e.g., two stage) compressors, and/or variable speed compressors. These and other features will be described below with reference to the drawings.

(30) Turning now to the drawings, FIG. 1 illustrates an embodiment of a heating, ventilation, and

air conditioning (HVAC) system for environmental management that employs one or more HVAC units in accordance with the present disclosure. As used herein, an HVAC system includes any number of components configured to enable regulation of parameters related to climate characteristics, such as temperature, humidity, air flow, pressure, air quality, and so forth. For example, an “HVAC system” as used herein is defined as conventionally understood and as further described herein. Components or parts of an “HVAC system” may include, but are not limited to, all, some of, or individual parts such as a heat exchanger, a heater, an air flow control device, such as a fan, a sensor configured to detect a climate characteristic or operating parameter, a filter, a control device configured to regulate operation of an HVAC system component, a component configured to enable regulation of climate characteristics, or a combination thereof. An “HVAC system” is a system configured to provide such functions as heating, cooling, ventilation, dehumidification, pressurization, refrigeration, filtration, or any combination thereof. The embodiments described herein may be utilized in a variety of applications to control climate characteristics, such as residential, commercial, industrial, transportation, or other applications where climate control is desired.

(31) In the illustrated embodiment, a building **10** is air conditioned by a system that includes an HVAC unit **12** with a reheat system in accordance with present embodiments. The building **10** may be a commercial structure or a residential structure. As shown, the HVAC unit **12** is disposed on the roof of the building **10**; however, the HVAC unit **12** may be located in other equipment rooms or areas adjacent the building **10**. The HVAC unit **12** may be a single package unit containing other equipment, such as a blower, integrated air handler, and/or auxiliary heating unit. In other embodiments, the HVAC unit **12** may be part of a split HVAC system, such as the system shown in FIG. **3**, which includes an outdoor HVAC unit **58** and an indoor HVAC unit **56**.

(32) The HVAC unit **12** is an air-cooled device that implements a refrigeration cycle to provide conditioned air to the building **10**. Specifically, the HVAC unit **12** may include one or more heat exchangers across which an air flow is passed to condition the air flow before the air flow is supplied to the building. In the illustrated embodiment, the HVAC unit **12** is a rooftop unit (RTU) that conditions a supply air stream, such as environmental air and/or a return air flow from the building **10**. After the HVAC unit **12** conditions the air, the air is supplied to the building **10** via ductwork **14** extending throughout the building **10** from the HVAC unit **12**. For example, the ductwork **14** may extend to various individual floors or other sections of the building **10**. In certain embodiments, the HVAC unit **12** may be a heat pump that provides both heating and cooling to the building with one refrigeration circuit configured to operate in different modes. In other embodiments, the HVAC unit **12** may include one or more working fluid circuits (e.g., refrigeration circuits) for cooling an air stream and a furnace for heating the air stream.

(33) A control device **16**, one type of which may be a thermostat, may be used to designate the temperature of the conditioned air. The control device **16** also may be used to control the flow of air through the ductwork **14**. For example, the control device **16** may be used to regulate operation of one or more components of the HVAC unit **12** or other components, such as dampers and fans, within the building **10** that may control flow of air through and/or from the ductwork **14**. In some embodiments, other devices may be included in the system, such as pressure and/or temperature transducers or switches that sense the temperatures and pressures of the supply air, return air, and so forth. Moreover, the control device **16** may include computer systems that are integrated with or separate from other building control or monitoring systems, and even systems that are remote from the building **10**.

(34) FIG. **2** is a perspective view of an embodiment of the HVAC unit **12**. In the illustrated embodiment, the HVAC unit **12** is a single package unit that may include one or more independent refrigeration circuits and components that are tested, charged, wired, piped, and ready for installation. The HVAC unit **12** may provide a variety of heating and/or cooling functions, such as cooling only, heating only, cooling with electric heat, cooling with dehumidification, cooling with

gas heat, or cooling with a heat pump. As described above, the HVAC unit **12** may directly cool and/or heat an air stream provided to the building **10** to condition a space in the building **10**.

(35) As shown in the illustrated embodiment of FIG. 2, a cabinet **24** encloses the HVAC unit **12** and provides structural support and protection to the internal components from environmental and other contaminants. In some embodiments, the cabinet **24** may be constructed of galvanized steel and insulated with aluminum foil faced insulation. Rails **26** may be joined to the bottom perimeter of the cabinet **24** and provide a foundation for the HVAC unit **12**. In certain embodiments, the rails **26** may provide access for a forklift and/or overhead rigging to facilitate installation and/or removal of the HVAC unit **12**. In some embodiments, the rails **26** may fit into “curbs” on the roof to enable the HVAC unit **12** to provide air to the ductwork **14** from the bottom of the HVAC unit **12** while blocking elements such as rain from leaking into the building **10**.

(36) The HVAC unit **12** includes heat exchangers **28** and **30** in fluid communication with one or more refrigeration circuits (e.g., working fluid circuits). Tubes within the heat exchangers **28** and **30** may circulate refrigerant, such as R-410A, through the heat exchangers **28** and **30**. The tubes may be of various types, such as multichannel tubes, conventional copper or aluminum tubing, and so forth. Together, the heat exchangers **28** and **30** may implement a thermal cycle in which the refrigerant undergoes phase changes and/or temperature changes as it flows through the heat exchangers **28** and **30** to produce heated and/or cooled air. For example, the heat exchanger **28** may function as a condenser where heat is released from the refrigerant to ambient air, and the heat exchanger **30** may function as an evaporator where the refrigerant absorbs heat to cool an air stream. In other embodiments, the HVAC unit **12** may operate in a heat pump mode where the roles of the heat exchangers **28** and **30** may be reversed. That is, the heat exchanger **28** may function as an evaporator and the heat exchanger **30** may function as a condenser. In further embodiments, the HVAC unit **12** may include a furnace for heating the air stream that is supplied to the building **10**. While the illustrated embodiment of FIG. 2 shows the HVAC unit **12** having two of the heat exchangers **28** and **30**, in other embodiments, the HVAC unit **12** may include one heat exchanger or more than two heat exchangers.

(37) The heat exchanger **30** is located within a compartment **31** that separates the heat exchanger **30** from the heat exchanger **28**. Fans **32** draw air from the environment through the heat exchanger **28**. Air may be heated and/or cooled as the air flows through the heat exchanger **28** before being released back to the environment surrounding the HVAC unit **12**. A blower assembly **34**, powered by a motor **36**, draws air through the heat exchanger **30** to heat or cool the air. The heated or cooled air may be directed to the building **10** by the ductwork **14**, which may be connected to the HVAC unit **12**. Before flowing through the heat exchanger **30**, the conditioned air flows through one or more filters **38** that may remove particulates and contaminants from the air. In certain embodiments, the filters **38** may be disposed on the air intake side of the heat exchanger **30** to prevent contaminants from contacting the heat exchanger **30**.

(38) The HVAC unit **12** also may include other equipment for implementing the thermal cycle. Compressors **42** increase the pressure and temperature of the refrigerant before the refrigerant enters the heat exchanger **28**. The compressors **42** may be any suitable type of compressors, such as scroll compressors, rotary compressors, screw compressors, or reciprocating compressors. In some embodiments, the compressors **42** may include a pair of hermetic direct drive compressors arranged in a dual stage configuration **44**. However, in other embodiments, any number of the compressors **42** may be provided to achieve various stages of heating and/or cooling. As may be appreciated, additional equipment and devices may be included in the HVAC unit **12**, such as a solid-core filter drier, a drain pan, a disconnect switch, an economizer, pressure switches, phase monitors, and humidity sensors, among other things.

(39) The HVAC unit **12** may receive power through a terminal block **46**. For example, a high voltage power source may be connected to the terminal block **46** to power the equipment. The operation of the HVAC unit **12** may be governed or regulated by a control board **48**. The control

board **48** may include control circuitry connected to a thermostat, sensors, and alarms. One or more of these components may be referred to herein separately or collectively as the control device **16**. The control circuitry may be configured to control operation of the equipment, provide alarms, and monitor safety switches. Wiring **49** may connect the control board **48** and the terminal block **46** to the equipment of the HVAC unit **12**.

(40) FIG. **3** illustrates a residential heating and cooling system **50**, also in accordance with present techniques. The residential heating and cooling system **50** may provide heated and cooled air to a residential structure, as well as provide outside air for ventilation and provide improved indoor air quality (IAQ) through devices such as ultraviolet lights and air filters. In the illustrated embodiment, the residential heating and cooling system **50** is a split HVAC system. In general, a residence **52** conditioned by a split HVAC system may include refrigerant conduits **54** (e.g., working fluid conduits) that operatively couple the indoor unit **56** to the outdoor unit **58**. The indoor unit **56** may be positioned in a utility room, an attic, a basement, and so forth. The outdoor unit **58** is typically situated adjacent to a side of residence **52** and is covered by a shroud to protect the system components and to prevent leaves and other debris or contaminants from entering the unit. The refrigerant conduits **54** transfer refrigerant between the indoor unit **56** and the outdoor unit **58**, typically transferring primarily liquid refrigerant in one direction and primarily vaporized refrigerant in an opposite direction.

(41) When the system shown in FIG. **3** is operating as an air conditioner, a heat exchanger **60** in the outdoor unit **58** serves as a condenser for re-condensing vaporized refrigerant flowing from the indoor unit **56** to the outdoor unit **58** via one of the refrigerant conduits **54**. In these applications, a heat exchanger **62** of the indoor unit functions as an evaporator. Specifically, the heat exchanger **62** receives liquid refrigerant, which may be expanded by an expansion device, and evaporates the refrigerant before returning it to the outdoor unit **58**.

(42) The outdoor unit **58** draws environmental air through the heat exchanger **60** using a fan **64** and expels the air above the outdoor unit **58**. When operating as an air conditioner, the air is heated by the heat exchanger **60** within the outdoor unit **58** and exits the unit at a temperature higher than it entered. The indoor unit **56** includes a blower or fan **66** that directs air through or across the indoor heat exchanger **62**, where the air is cooled when the system is operating in air conditioning mode. Thereafter, the air is passed through ductwork **68** that directs the air to the residence **52**. The overall system operates to maintain a desired temperature as set by a system controller. When the temperature sensed inside the residence **52** is higher than the set point on the thermostat, or the set point plus a small amount, the residential heating and cooling system **50** may become operative to refrigerate additional air for circulation through the residence **52**. When the temperature reaches the set point, or the set point minus a small amount, the residential heating and cooling system **50** may stop the refrigeration cycle temporarily. The outdoor unit **58** may include a reheat system in accordance with present embodiments.

(43) The residential heating and cooling system **50** may also operate as a heat pump. When operating as a heat pump, the roles of heat exchangers **60** and **62** are reversed. That is, the heat exchanger **60** of the outdoor unit **58** will serve as an evaporator to evaporate refrigerant and thereby cool air entering the outdoor unit **58** as the air passes over the outdoor heat exchanger **60**. The indoor heat exchanger **62** will receive a stream of air blown over it and will heat the air by condensing the refrigerant.

(44) In some embodiments, the indoor unit **56** may include a furnace system **70**. For example, the indoor unit **56** may include the furnace system **70** when the residential heating and cooling system **50** is not configured to operate as a heat pump. The furnace system **70** may include a burner assembly and heat exchanger, among other components, inside the indoor unit **56**. Fuel is provided to the burner assembly of the furnace **70** where it is mixed with air and combusted to form combustion products. The combustion products may pass through tubes or piping in a heat exchanger, separate from heat exchanger **62**, such that air directed by the blower **66** passes over the

tubes or pipes and extracts heat from the combustion products. The heated air may then be routed from the furnace system **70** to the ductwork **68** for heating the residence **52**.

(45) FIG. **4** is an embodiment of a vapor compression system **72** that can be used in any of the systems described above. The vapor compression system **72** may circulate a working fluid (e.g., refrigerant) through a circuit starting with a compressor **74**. The circuit may also include a condenser **76**, an expansion valve(s) or device(s) **78**, and an evaporator **80**. The vapor compression system **72** may further include a control panel **82** that has an analog to digital (A/D) converter **84**, a microprocessor **86**, a non-volatile memory **88**, and/or an interface board **90**. The control panel **82** and its components may function to regulate operation of the vapor compression system **72** based on feedback from an operator, from sensors of the vapor compression system **72** that detect operating conditions, and so forth.

(46) In some embodiments, the vapor compression system **72** may use one or more of a variable speed drive (VSDs) **92**, a motor **94**, the compressor **74**, the condenser **76**, the expansion valve or device **78**, and/or the evaporator **80**. The motor **94** may drive the compressor **74** and may be powered by the variable speed drive (VSD) **92**. The VSD **92** receives alternating current (AC) power having a particular fixed line voltage and fixed line frequency from an AC power source, and provides power having a variable voltage and frequency to the motor **94**. In other embodiments, the motor **94** may be powered directly from an AC or direct current (DC) power source. The motor **94** may include any type of electric motor that can be powered by a VSD or directly from an AC or DC power source, such as a switched reluctance motor, an induction motor, an electronically commutated permanent magnet motor, or another suitable motor.

(47) The compressor **74** compresses a refrigerant vapor and delivers the vapor to the condenser **76** through a discharge passage. In some embodiments, the compressor **74** may be a centrifugal compressor. The refrigerant vapor delivered by the compressor **74** to the condenser **76** may transfer heat to a fluid passing across the condenser **76**, such as ambient or environmental air **96**. The refrigerant vapor may condense to a refrigerant liquid in the condenser **76** as a result of thermal heat transfer with the environmental air **96**. The liquid refrigerant from the condenser **76** may flow through the expansion device **78** to the evaporator **80**.

(48) The liquid refrigerant delivered to the evaporator **80** may absorb heat from another air stream, such as a supply air stream **98** provided to the building **10** or the residence **52**. For example, the supply air stream **98** may include ambient or environmental air, return air from a building, or a combination of the two. The liquid refrigerant in the evaporator **80** may undergo a phase change from the liquid refrigerant to a refrigerant vapor. In this manner, the evaporator **80** may reduce the temperature of the supply air stream **98** via thermal heat transfer with the refrigerant. Thereafter, the vapor refrigerant exits the evaporator **80** and returns to the compressor **74** by a suction line to complete the cycle.

(49) In some embodiments, the vapor compression system **72** may further include a reheat coil. In the illustrated embodiment, the reheat coil is represented as part of the evaporator **80**. The reheat coil is positioned downstream of the evaporator heat exchanger relative to the supply air stream **98** and may reheat the supply air stream **98** when the supply air stream **98** is overcooled to remove humidity from the supply air stream **98** before the supply air stream **98** is directed to the building **10** or the residence **52**.

(50) It should be appreciated that any of the features described herein may be incorporated with the HVAC unit **12**, the residential heating and cooling system **50**, or other HVAC systems.

Additionally, while the features disclosed herein are described in the context of embodiments that directly heat and cool a supply air stream provided to a building or other load, embodiments of the present disclosure may be applicable to other HVAC systems as well. For example, the features described herein may be applied to mechanical cooling systems, free cooling systems, chiller systems, or other heat pump or refrigeration applications.

(51) As briefly discussed above, embodiments of the present disclosure are directed to an HVAC

system having an improved heat pump system. To provide context for the following discussion, FIG. 5 is a schematic of an embodiment of a portion of an HVAC system **100** that includes a heat pump **102** (e.g., a heat pump system, reverse-cycle heat pump) in accordance with present embodiments. The heat pump **102** may include a portion of or all of the components of the vapor compression system **72** discussed above. The heat pump **102** includes a first heat exchanger **104** and a second heat exchanger **106** that are fluidly coupled to one another via a refrigerant circuit **108** (e.g., one or more conduits, working fluid circuit). The first heat exchanger **104** may be in thermal communication with (e.g., fluidly coupled to) a thermal load **110** (e.g., a room, space, and/or device) serviced by the heat pump **102**, and the second heat exchanger **106** may be in thermal communication with an ambient environment **112** (e.g., the atmosphere) surrounding the HVAC system **100**.

(52) In some embodiments, a first fan **116** may direct a first air flow across the first heat exchanger **104** to facilitate heat exchange between refrigerant within the first heat exchanger **104** and the thermal load **110**, while a second fan **118** may direct a second air flow across the second heat exchanger **106** to facilitate heat exchange between refrigerant within the second heat exchanger **106** and the ambient environment. An expansion device **120** (e.g., an electronic expansion valve) may be disposed along the refrigerant circuit **108** between the first heat exchanger **104** and the second heat exchanger **106** and may be configured to regulate (e.g., throttle) a refrigerant flow and/or a refrigerant pressure differential between the first and second heat exchangers **104**, **106**.

(53) The heat pump **102** also includes a compressor system **130** disposed along the refrigerant circuit **108**. The compressor system **130** includes a plurality of compressors **132**, such as a first compressor **134** and a second compressor **136**, which, as discussed below, are each configured to direct refrigerant flow through the first heat exchanger **104**, the second heat exchanger **106**, and remaining components (e.g., the expansion device **120**) that may be fluidly coupled to the refrigerant circuit **108**. Although the compressor system **130** is shown as having two compressors **132** in the illustrated embodiment, the compressor system **130** may include any suitable quantity of compressors **132**, such as two, three, four, five, six, or more than six compressors **132**. In any case, at least a subset of the compressors **132** may be fluidly coupled to one another in a parallel configuration or a parallel arrangement (e.g., relative to a flow of refrigerant through the compressors **132**). For example, the compressor system **130** may include a suction conduit **140** and a discharge conduit **142**, and the first and second compressors **134**, **136** may be fluidly coupled to the suction conduit **140** and the discharge conduit **142** in a parallel flow configuration with respect to one another. In this way, the first compressor **134** and the second compressor **136** may each be operable to draw (e.g., intake) a refrigerant flow from the suction conduit **140** and discharge (e.g., output) the refrigerant flow through the discharge conduit **142**. The compressor system **130** may include one or more valves **148** (e.g., check valves, solenoid valve, modulating valves) that are operable to enable the first compressor **134**, the second compressor **136**, both the first and second compressors **134**, **136**, and/or additional compressors **132** to direct refrigerant through the refrigerant circuit **108**. In particular, the one or more valves **148** may be controlled (e.g., adjusted) to adjust refrigerant flow through one or more of the compressors **132**.

(54) The compressor system **130** may be fluidly coupled to a remainder of the refrigerant circuit **108** via a reversing valve **150** (e.g., a switch-over valve). In particular, the reversing valve **150** may include a first port **152** that is fluidly coupled to the suction conduit **140**, a second port **154** that is fluidly coupled to the discharge conduit **142**, a third port **156** that is fluidly coupled to a first conduit portion **158** of the refrigerant circuit **108** extending to the first heat exchanger **104**, and a fourth port **160** that is fluidly coupled to a second conduit portion **162** of the refrigerant circuit **108** extending to the second heat exchanger **106**.

(55) The reversing valve **150** is configured to transition between a first configuration **164**, in which the reversing valve **150** fluidly couples the first port **152** and the third port **156** and fluidly couples the second port **154** and the fourth port **160**, and a second configuration **170**, illustrated in FIG. 6,

in which the reversing valve **150** fluidly couples the first port **152** and the fourth port **160** and fluidly couples the second port **154** and the third port **156**. Accordingly, in the first configuration **164**, the reversing valve **150** enables the compressors **132** to receive a flow of refrigerant from the first heat exchanger **104** and to discharge a flow of refrigerant to the second heat exchanger **106**. Conversely, in the second configuration **170**, the reversing valve **150** enables the compressors **132** to receive a flow of refrigerant from the second heat exchanger **106** and to discharge a flow of refrigerant to the first heat exchanger **104**. In this way, while in the first configuration **164**, the reversing valve **150** enables the heat pump **102** to operate in a cooling mode, in which the first heat exchanger **104** absorbs thermal energy from the thermal load **110** to cool the thermal load **110**, and the second heat exchanger **106** rejects the absorbed thermal energy (e.g., absorbed from the thermal load **110**) to the ambient environment **112**. Further, while in the second configuration **170**, the reversing valve **150** enables the heat pump **102** to operate in a heating mode, in which the second heat exchanger **106** absorbs thermal energy from the ambient environment **112**, and the first heat exchanger **104** rejects the absorbed thermal energy (e.g., absorbed from the ambient environment **112**) to the thermal load **110** to heat the thermal load **110**. As such, while the reversing valve **150** is in the first configuration **164**, the compressor system **130** may direct a refrigerant flow along at least a portion of the refrigerant circuit **108** in a first flow direction **172**. While the reversing valve **150** is in the second configuration **170**, the compressor system **130** may direct a refrigerant flow along at least a portion of the refrigerant circuit **108** in a second flow direction **174**, opposite the first flow direction **172**. For clarity, the heat pump **102** is shown configured for operation in a cooling mode in the illustrated embodiment of FIG. 5. Moreover, FIG. 6 is a schematic of an embodiment of a portion of the HVAC system **100**, illustrating the heat pump **102** configured for operation in a heating mode.

(56) The present discussion continues with reference to FIG. 5. The HVAC system **100** may include a controller **180** (e.g., a control system, a thermostat, a control panel) that is communicatively coupled to one or more components of the heat pump **102** and is configured to monitor, adjust, and/or otherwise control operation of the components of the heat pump **102**. For example, one or more control transfer devices, such as wires, cables, wireless communication devices, and the like, may communicatively couple the compressors **132**, the reversing valve **150**, the expansion device **120**, the first and/or second fans **116**, **118**, the control device **16** (e.g., a thermostat), and/or any other suitable components of the HVAC system **100** to the controller **180**. That is, the compressors **132**, the reversing valve **150**, the expansion device **120**, the first and/or second fans **116**, **118**, and/or the control device **16** may each have a communication component that facilitates wired or wireless (e.g., via a network) communication with the controller **180**. In some embodiments, the communication components may include a network interface that enables the components of the HVAC system **100** to communicate via various protocols such as EtherNet/IP, ControlNet, DeviceNet, or any other communication network protocol. Alternatively, the communication component may enable the components of the HVAC system **100** to communicate via mobile telecommunications technology, Bluetooth®, near-field communications technology, and the like. As such, the compressors **132**, the reversing valve **150**, the expansion device **120**, the first and/or second fans **116**, **118**, and/or the control device **16** may wirelessly communicate data between each other.

(57) In some embodiments, the controller **180** may include a portion or all of the control panel **82** illustrated in FIG. 4 or may be another suitable controller included in the HVAC system **100**. In any case, the controller **180** may be configured to control components of the HVAC system **100** in accordance with the techniques discussed herein. The controller **180** includes processing circuitry **182** (e.g., one or more processors), such as a microprocessor, which may execute software for controlling the components of the HVAC system **100**. The processing circuitry **182** may include multiple microprocessors, one or more “general-purpose” microprocessors, one or more special-purpose microprocessors, and/or one or more application specific integrated circuits (ASICs), or

some combination thereof. For example, the processing circuitry **182** may include one or more reduced instruction set (RISC) processors.

(58) The controller **180** may also include a memory device **184** (e.g., a memory) that may store information such as instructions, control software, look up tables, configuration data, etc. The memory device **184** may include a volatile memory, such as random access memory (RAM), and/or a nonvolatile memory, such as read-only memory (ROM). The memory device **184** may store a variety of information and may be used for various purposes. For example, the memory device **184** may store processor-executable instructions including firmware or software for the processing circuitry **182** execute, such as instructions for controlling components of the HVAC system **100**. In some embodiments, the memory device **184** is a tangible, non-transitory, machine-readable-medium that may store machine-readable instructions for the processing circuitry **182** to execute. The memory device **184** may include ROM, flash memory, a hard drive, or any other suitable optical, magnetic, or solid-state storage medium, or a combination thereof. The memory device **184** may store data, instructions, and any other suitable data.

(59) To facilitate the following discussion, FIG. 7 is flow diagram of an embodiment of a process **200** for controlling the heat pump **102** in accordance with the presently disclosed techniques. FIG. 7 will be referred to concurrently with FIGS. 5 and 6 throughout the following discussion. It should be noted that the steps of the process **200** discussed below may be performed in any suitable order and are not limited to the order shown in the illustrated embodiment of FIG. 7. Moreover, it should be noted that additional steps of the process **200** may be performed, and certain steps of the process **200** may be omitted. In some embodiments, the process **200** may be executed by the processing circuitry **182** of the controller **180** and/or any other suitable controller of the HVAC system **100**. The process **200** may be stored on, for example, the memory **88** or the memory device **184**.

(60) The process **200** may begin with receiving a call for cooling or heating, as indicated by block **202**. For example, the controller **180** may receive a call (e.g., a control instruction) from the control device **16** or another suitable controller instructing the controller **180** to operate the heat pump **102** in the cooling mode to cool the thermal load **110** or to the heating mode to heat the thermal load **110**. In response to receiving the call to operate the heat pump **102** in the cooling mode, the controller **180** may instruct the reversing valve **150** to transition to the first configuration **164** illustrated in FIG. 5. Conversely, in response to receiving the call to operate the heat pump **102** in the heating mode, the controller **180** may instruct the reversing valve **150** to transition to the second configuration **170** illustrated in FIG. 6.

(61) The process **200** includes selecting one of the compressors **132** or a combination of the compressors **132** that enables improved operation of the heat pump **102** in either the heating mode or the cooling mode, as indicated by block **204**. For example, in some embodiments, the first compressor **134** may include operational characteristics that enable the first compressor **134** to more efficiently direct refrigerant through the refrigerant circuit **108** during operation of the heat pump **102** in the cooling mode (e.g., as compared to operation of the second compressor **136** to direct refrigerant through the refrigerant circuit **108** in the cooling mode). The second compressor **136** may include operational characteristics that enable the second compressor **136** to more efficiently direct refrigerant through the refrigerant circuit **108** during operation of the heat pump **102** in the heating mode (e.g., as compared to operation of the first compressor **134** to direct refrigerant through the refrigerant circuit **108** in the heating mode). The operational characteristics of the compressors **132** may include respective volume indices or compression ratios of the compressors **132**, respective capacities or displacements of the compressors **132** (e.g., a volume of fluid ingested by the compressor **132** per revolution of the compressor **132**), and/or other suitable parameters of the compressors **132**. As an example, the first compressor **134** may have a first BVR or volume index (e.g., 2.1, 2.2, 2.3, 2.4, 2.5, 2.6), and the second compressor **136** may have a second BVR or volume index (e.g., 3.6 3.7, 3.8, 3.9, 4.0, 4.1, 4.2) that is greater than the first BVR. In certain embodiments, the operational characteristics of the first compressor **134** and/or the

second compressor **136** may be selected based on a climatic region (e.g., a geographical location) in which the heat pump **102** is implemented and operated. For clarity, the first compressor **134** may also be referred to herein as a cooling compressor **134**, and the second compressor **136** may also be referred to as a heating compressor **136**.

(62) The controller **180** may, at block **204**, select a particular one or combination of the compressors **132** for operation based on whether a call for cooling or a call for heating is received at the block **202** and based on the operational characteristics of the compressors **132** and/or operational parameters of the HVAC system **100**. In particular, the controller **180** may designate a particular one or combination of the compressors **132** for operation in a manner that enhances an overall operational efficiency of the HVAC system **100** while the heat pump **102** operates in either the cooling mode or the heating mode, and may proceed to activate the selected compressors **132** or combination of compressors **132**, as indicated by block **206**.

(63) For example, in some embodiments, in response to receiving a call for cooling at block **202** (e.g., from the control device **16**), the controller **180** may send control signals to transition the reversing valve **150** to the first configuration **164** shown in FIG. 5, to activate the cooling compressor **134**, and to transition or retain the heating compressor **136** in an idle (e.g., inactive) state. As such, the cooling compressor **134** may direct refrigerant along the refrigerant circuit **108** (e.g., in the first flow direction **172**) to operate the heat pump **102** in the cooling mode, while the heating compressor **136** may be inactive and may not operate to circulate refrigerant along the refrigerant circuit **108**. In response to receiving a call for heating at block **202** (e.g., from the control device **16**), the controller **180** may send control signals to transition the reversing valve **150** to the second configuration **170** shown in FIG. 6, to activate the heating compressor **136**, and to transition or retain the cooling compressor **134** in an idle (e.g., inactive) state. Accordingly, the heating compressor **136** may direct refrigerant along the refrigerant circuit **108** in the second flow direction **174** to operate the heat pump **102** in the heating mode while the cooling compressor **134** may be inactive and may not operate to circulate refrigerant along the refrigerant circuit **108**.

(64) In some embodiments, the controller **180** may operate both the cooling compressor **134** and the heating compressor **136** in conjunction with one another and in a manner that further improves the operational efficiency of the heat pump **102** in the cooling mode or the heating mode (e.g., by reducing electrical resistance heat that may be involved to meet relatively large heating loads). To better illustrate and to facilitate the following discussion, FIG. 8 is flow diagram of an embodiment of a process **240** for controlling one or more of the compressors **132** to enhance an operational efficiency of the heat pump **102** in the heating mode. However, it should be understood that the process **240** may be implemented in a similar manner to improve the operational efficiency of the heat pump **102** in the cooling mode. FIG. 8 will be referred to concurrently with FIGS. 5 and 6 throughout the following discussion. It should be noted that the steps of the process **240** discussed below may be performed in any suitable order and are not limited to the order shown in the illustrated embodiment of FIG. 8. Moreover, it should be noted that additional steps of the process **240** may be performed, and certain steps of the process **240** may be omitted. In some embodiments, the process **240** may be executed by the processing circuitry **182** of the controller **180** and/or any other suitable controller of the HVAC system **100**. The process **240** may be stored on, for example, the memory **88** or the memory device **184**.

(65) The process **240** includes determining (e.g., via the controller **180**) whether there is an outstanding call for heating (e.g., from the control device **16**), as indicated by block **242**. In response to determining (e.g., based on signals received from the control device **16**) that a call for heating exists, the controller **180** may determine, at block **244**, whether a call for multi-compressor or increased (e.g., second) stage compressor operation is desired. The controller **180** may determine whether the call for multi-compressor or increased stage compressor operation exists via evaluation of one or more operating parameters of the HVAC system **100**, as discussed below.

(66) As an example, in some embodiments, the controller **180** may determine that a call for multi-

compressor or increased compressor stage operation exists in response to receiving feedback (e.g., data) from the control device **16** that a temperature of or within the thermal load **110** (e.g., an air temperature in the thermal load **110**, a detected temperature, a measured temperature) deviates from a target temperature set point (e.g., set point temperature) for the thermal load **110** by a threshold amount and/or by a threshold percentage (e.g., a threshold value). In other words, in the controller **180** may determine that a call for multi-compressor exists in response to a determination that a temperature differential between the temperature of or within the thermal load **110** and the target temperature set point is equal to or greater than the threshold amount. In some embodiments, the controller **180** may receive data indicative the temperature of or within the thermal load **110** from a sensor, such as a room air sensor or a return air sensor, and the controller **180** may receive data indicative of target temperature set point for the thermal load **110** from the control device **16** (e.g., a thermostat). Based on the data, the controller **180** may determine a temperature differential between the temperature of the thermal load **110** and the set point temperature and may compare the temperature differential to the threshold amount.

(67) The controller **180** may determine that multi-compressor or increased compressor stage operation is desired in response to determining that a heating demand of the thermal load **110** is relatively high, such as when the temperature of or within the thermal load **110** is below the target temperature set point for the thermal load **110** by the threshold amount. Additionally or alternatively, the controller **180** may determine that multi-compressor or increased compressor stage operation is desired based on a time of day at which the call for heating is received at block **242**, based on an occupancy within the thermal load **110** at the time the call for heating is received, based on ambient atmospheric conditions surrounding the HVAC system **100** at the time the call for heating is received, based on a suction pressure of the compressor system **130**, based on a discharge pressure of the compressor system **130**, based on an operational speed of the fans **116** and/or **118**, or any combination thereof.

(68) In any case, in response to determining that a call for heating exists at block **242** and that multi-stage compressor operation is not desired at block **244**, the controller **180** may send instructions (e.g., control signals) to operate the heating compressor **136** and may transition and/or retain the cooling compressor **134** in an idle state, as indicated by block **246**. In response to determining that a call for heating exists at block **242** and that multi-stage compressor operation is desired at block **244**, the controller **180** may send instructions to operate both the heating compressor **136** and the cooling compressor **134**, as indicated by block **248**.

(69) In some embodiments, the controller **180** may determine a demand level of the thermal load **110** at block **244**. The controller **180** may, based on the determined demand level of the thermal load **110**, determine whether to operate one or multiple of the compressors **132** and, in embodiments where a call for multi-compressor operation is determined, stage sequential operation of the cooling compressor **134**, the heating compressor **136**, and/or additional compressors **132** that may be included in the compressor system **130** during execution of the block **246**. For example, in some embodiments, the controller **180** may determine that a mild heating demand level exists for the thermal load **110** in response to determining that a temperature of or within the thermal load **110** is below a target temperature set point of the thermal load **110** by a first threshold amount or percentage. The controller **180** may determine that a moderate heating demand level exists for the thermal load **110** in response to determining that the temperature of or within the thermal load **110** is below the target temperature set point of the thermal load **110** by a second threshold amount or percentage that is greater than the first threshold amount or percentage. Further, the controller **180** may determine that a high heating demand level exists for the thermal load **110** in response to determining that the temperature of or within the thermal load **110** is below the target temperature set point of the thermal load **110** by a third threshold amount or percentage that is greater than the second threshold amount or percentage.

(70) In response to determining that a mild heating demand level exists for the thermal load **110** at

block **244**, the controller **180** may send instructions to activate the cooling compressor **134** and to deactivate the heating compressor **136** (and/or retain the heating compressor **136** in an inactive state). In some embodiments, the cooling compressor **134** may have a capacity (e.g., operating output) that is less than a capacity of the heating compressor **136**. However, the relatively low capacity of the cooling compressor **134** may still be sufficient to enable the heat pump **102** to adequately satisfy the mild heating demand of the thermal load **110**, while also enabling the heat pump **102** to operate at an improved efficiency level (e.g., using less electrical power), as compared to operating the heating compressor **136** to circulate refrigerant through the refrigerant circuit **108** to satisfy the mild heating demand. For example, during operation to satisfy the mild heating demand, a pressure ratio across the compressor system **130** may be relatively low, and the cooling compressor **134**, which may be configured to operate more efficiently at lower pressure ratios than the heating compressor **136**, may operate with improved efficiency to satisfy the mild heating demand.

(71) In response to determining that a moderate heating demand level exists for the thermal load **110** at block **244**, the controller **180** may send instructions to deactivate the cooling compressor **134** (and/or retain the cooling compressor **134** in an inactive state) and to activate the heating compressor **136**, which may operate more efficiently than the cooling compressor **134** at higher pressure ratios associated with the moderate heating demand. As such, the controller **180** may operate the heating compressor **136** in a manner that enables the heat pump **102** to adequately satisfy the moderate heating demand level of the thermal load **110** with improved efficiency. Further, in response to determining that a high heating demand level exists for the thermal load **110** at block **244**, the controller **180** may send instructions activate both the cooling compressor **134** and the heating compressor **136** to increase an overall output (e.g., capacity) of the compressor system **130** and enable the heat pump **102** to adequately satisfy the relatively high heating demand level of the thermal load **110**.

(72) In embodiments of the compressor system **130** having more than two compressors **132**, the controller **180** may selectively activate or deactivate any one or combination of the compressors **132** based on a current (e.g., existing, updated) heating or cooling demand of the thermal load **110** and/or based on one or more measured operational parameters of the HVAC system **100** (e.g., measured ambient or outdoor air temperatures) to enable the heat pump **102** to adequately satisfy the heating or cooling demand of the thermal load **110** with improved efficiency. For example, the controller **180** may be configured to sequentially activate one, two, three, four, five, six, or more than six compressors **132** of the compressor system **130** based on the current heating or cooling demand of the thermal load **110** and/or based on one or more measured operational parameters of the HVAC system **100** to enable the heat pump **102** to adequately satisfy the heating or cooling demand of the thermal load **110** with improved efficiency. In some embodiments, multiple compressors **132** may be sequentially activated based on respective operating efficiencies of the compressors **132** (e.g., a most efficient non-operating compressor **132** is activated before other non-operating compressors **132** are activated). Moreover, in certain embodiments, one or more of the compressors **132** may include multi-stage compressors **132** or variable speed compressors **132**. In such embodiments, the controller **180** may be configured to selectively adjust stages of one or more of the compressors **132** and/or speeds of one or more of the compressors **132** in a manner that enables the heat pump **102** to adequately satisfy the cooling or heating demand of the thermal load **110** with improved efficiency. The controller **180** may be configured to adjust operation of the compressors **132** in accordance with the aforementioned techniques based on sensor feedback (e.g., ambient temperature sensor feedback), control instructions received from other control devices of the HVAC system **100**, user input provided via a user interface **258** of the HVAC system **100** that is communicatively coupled to the controller **180**, and/or based on other suitable control instructions received by the controller **180**.

(73) FIG. **9** is flow diagram of another embodiment of the process **240**, referred to herein as a

process **260**. FIGS. **5**, **6**, and **9** are discussed concurrently below. In the illustrated embodiment of the process **260**, in response to determining that a call for heating exists at block **242** and that multi-stage compressor operation is not desired at block **244**, the controller **180** may determine whether an ambient temperature of the ambient environment **112** is less than an ambient temperature set point (e.g., set point value, threshold value), as indicated by block **262**. For example, in some embodiments, the controller **180** may be communicatively coupled to a temperature sensor **264**, as shown in FIG. **5**, or a plurality of temperature sensors, configured to measure a temperature of the ambient environment **112**. In response to receiving feedback from the temperature sensor **264**, the controller **180** may compare the temperature of the ambient environment **112** to the ambient temperature set point (e.g., which may be stored in the memory device **184**). The controller **180** may receive the ambient temperature set point via a user input provided via the user interface **258** of the HVAC system **100**, for example.

(74) In response to a determination that the ambient temperature is greater than the ambient temperature set point at block **262** (e.g., associated with lower pressure ratios across the compressor system **130**), the controller **180** may send instructions to operate the cooling compressor **134** and to deactivate the heating compressor **136** (and/or retain the heating compressor **136** in an inactive state), as indicated by block **266**. In some embodiments, while the temperature of the ambient environment **112** is greater than the ambient temperature set point, the cooling compressor **134** may have a capacity (e.g., an operating output) that is sufficient to enable the heat pump **102** to adequately satisfy the heating demand of the thermal load **110**, while utilizing a relatively low amount of power (e.g., as compared to an amount of power consumed by the heating compressor **136**, which may have a greater capacity than the cooling compressor **134**). Conversely, in response to a determination that the ambient temperature is equal to or less than the ambient temperature set point at block **262** (e.g., associated with higher pressure ratios across the compressor system **130**), the controller **180** may send instructions to operate the heating compressor **136** and to deactivate the cooling compressor **134** (and/or retain the cooling compressor **134** in an inactive state), as indicated by block **246**. Additionally or alternatively, the controller **180** may implement the process **260** to operate more than two compressors **132**, to operate one or more multi-stage compressors **132**, and/or to operate one or more variable speed compressors **132** in accordance with the aforementioned techniques.

(75) FIG. **10** is flow diagram of another embodiment of the process **240**, referred to herein as a process **270**. FIGS. **5**, **6**, and **10** are discussed concurrently below. In the illustrated embodiment of the process **270**, in response to determining that a call for heating exists at block **242** and that multi-stage compressor operation is not desired at block **244**, the controller **180** may determine whether a coil temperature (e.g., outdoor coil temperature) of the second heat exchanger **106** (e.g., an outdoor heat exchanger or coil) is less than a coil temperature set point (e.g., outdoor coil temperature set point, set point value, threshold value) for the second heat exchanger **106**, as indicated by block **272**. For example, in some embodiments, the controller **180** may be communicatively coupled to a coil temperature sensor **274**, as shown in FIG. **5**, such as a temperature probe, that is configured to measure a coil temperature of the second heat exchanger **106**. The coil temperature of the second heat exchanger **106** may be indicative of a surface temperature of a portion of the second heat exchanger **106**, a temperature of refrigerant entering into or discharging from the second heat exchanger **106**, or another suitable (e.g., representative) temperature of the second heat exchanger **106**. In response to receiving feedback from the coil temperature sensor **274**, the controller **180** may compare the coil temperature of the second heat exchanger **106** to the coil temperature set point (e.g., which may be stored in the memory device **184**). The controller **180** may receive the coil temperature set point via a user input provided at the user interface **258**, for example.

(76) In response to a determination that the coil temperature of the second heat exchanger **106** is greater than the coil temperature set point at the block **272**, the controller **180** may send instructions

to operate the cooling compressor **134** and to deactivate the heating compressor **136** (and/or retain the heating compressor **136** in an inactive state), as indicated by block **266**. In some embodiments, while the coil temperature of the second heat exchanger **106** is greater than the coil temperature set point, the cooling compressor **134** may have a capacity (e.g., an operating output) that is sufficient to enable the heat pump **102** to adequately satisfy the heating demand of the thermal load **110**, while utilizing a relatively low amount of power (e.g., as compared to an amount of power consumed by the heating compressor **136**, which may have a greater capacity than the cooling compressor **134**). Conversely, in response to a determination that the coil temperature of the second heat exchanger **106** is equal to or less than the coil temperature set point at block **272**, the controller **180** may send instructions to operate the heating compressor **136** and to deactivate the cooling compressor **134** (and/or retain the cooling compressor **134** in an inactive state), as indicated by block **246**. Additionally or alternatively, the controller **180** may implement the process **270** to operate more than two compressors **132**, to operate one or more multi-stage compressors **132**, and/or to operate one or more variable speed compressors **132** in accordance with the aforementioned techniques.

(77) FIG. **11** is flow diagram of another embodiment of the process **240**, referred to herein as a process **280**. FIGS. **5**, **6**, and **11** are discussed concurrently below. In the illustrated embodiment of the process **280**, in response to determining that a call for heating exists at block **242** and that multi-stage compressor operation is not desired at block **244**, the controller **180** may determine whether a compressor suction pressure is less than a suction pressure set point (e.g., set point value, threshold value), as indicated by block **282**. For example, in some embodiments, the controller **180** may be communicatively coupled to a pressure sensor **284**, as shown in FIG. **5**, configured to measure a pressure (e.g., a suction pressure) along a suitable portion of the refrigerant circuit **108** (e.g., along the suction conduit **140**) or at a corresponding inlet of one of the compressors **132**. In response to receiving feedback from the pressure sensor **284**, the controller **180** may compare the suction pressure to the suction pressure set point (e.g., which may be stored in the memory device **184**). The controller **180** may receive the suction pressure set point via a user input provided at the user interface **258**, for example.

(78) In response to a determination that the suction pressure is greater than the suction pressure set point at block **282**, the controller **180** may send instructions to operate the cooling compressor **134** and to deactivate the heating compressor **136** (and/or retain the heating compressor **136** in an inactive state), as indicated by block **266**. Conversely, in response to a determination that the suction pressure is equal to or less than the suction pressure set point at block **282**, the controller **180** may send instructions to operate the heating compressor **136** and to deactivate the cooling compressor **134** (and/or retain the cooling compressor **134** in an inactive state), as indicated by block **246**. Additionally or alternatively, the controller **180** may implement the process **280** to operate more than two compressors **132**, to operate one or more multi-stage compressors **132**, and/or to operate one or more variable speed compressors **132** in accordance with the aforementioned techniques.

(79) FIG. **12** is flow diagram of another embodiment of the process **240**, referred to herein as a process **290**. FIGS. **5**, **6**, and **12** are discussed concurrently below. In the illustrated embodiment of the process **290**, in response to determining that a call for heating exists at block **242** and that multi-stage compressor operation is not desired at block **244**, the controller **180** may determine whether a suction saturation temperature of refrigerant along a portion of the refrigerant circuit **108** is less than a suction saturation temperature set point (e.g., threshold value, set point value), as indicated by block **292**. For example, in some embodiments, the controller **180** may be communicatively coupled to a suction saturation temperature sensor **294**, as shown in FIG. **6**, configured to measure a suction saturation temperature of refrigerant along a suitable portion of the refrigerant circuit **108**. In response to receiving feedback from the suction saturation temperature sensor **294**, the controller **180** may compare the suction saturation temperature to the suction

saturation temperature set point (e.g., which may be stored in the memory device **184**). The controller **180** may receive the suction saturation temperature set point via a user input provided at the user interface **258**, for example. It should be appreciated that, additionally or alternatively, the controller **180** may calculate the suction saturation temperature based on a suction pressure measurement (e.g., as determined via feedback provided by a suitable sensor) and thermodynamic property relations of the working fluid (e.g., refrigerant) implemented in the refrigerant circuit **108**.

(80) In any case, in response to a determination that the suction saturation temperature is greater than the suction saturation temperature set point at block **292**, the controller **180** may send instructions to operate the cooling compressor **134** and to deactivate the heating compressor **136** (and/or retain the heating compressor **136** in an inactive state), as indicated by block **266**. Conversely, in response to a determination that the suction saturation temperature is equal to or less than the suction saturation temperature set point at block **292**, the controller **180** may send instructions to operate the heating compressor **136** and to deactivate the cooling compressor **134** (and/or retain the cooling compressor **134** in an inactive state), as indicated by block **246**. Additionally or alternatively, the controller **180** may implement the process **280** to operate more than two compressors **132**, to operate one or more multi-stage compressors **132**, and/or to operate one or more variable speed compressors **132** in accordance with the aforementioned techniques.

(81) FIG. **13** is flow diagram of another embodiment of the process **240**, referred to herein as a process **296**. FIGS. **5**, **6**, and **13** are discussed concurrently below. In the illustrated embodiment of the process **296**, in response to determining that a call for heating exists at block **242** and that multi-stage compressor operation is not desired at block **244**, the controller **180** may determine whether a current isentropic efficiency of the heating compressor **136** is greater than a current isentropic efficiency of the cooling compressor **134**, as indicated by block **297**. For example, in some embodiments, the controller **180** may be communicatively coupled to an inlet pressure sensor **298**, as shown in FIG. **5**, that is disposed along the suction conduit **140** and is configured to measure a common inlet pressure (e.g., of the compressors **132**) along the suction conduit **140**. Further, the controller **180** may be communicatively coupled to an outlet pressure sensor **299** that is disposed along the discharge conduit **142** and is configured to measure a common outlet pressure (e.g., of the compressors **132**) along the discharge conduit **142**. The controller **180** may utilize feedback from the inlet and outlet pressure sensors **298**, **299** to calculate (e.g., using algorithms and/or lookup tables stored in the memory device **184**) a theoretical isentropic efficiency of each of the compressors **132** at the current operational state of the refrigerant circuit **108** (e.g., of the compressors **132**). Additionally or alternatively, the controller **180** may calculate or estimate the isentropic efficiency and/or pressure ratios of one or more of the compressors **132** based on sensor feedback indicative of an entering air temperature of the first heat exchanger **104** (e.g., an indoor coil) and an entering air temperature of the second heat exchanger **106** (e.g., an outdoor coil).

(82) In response to a determination that the calculated isentropic efficiency of the heating compressor **136** is less than or equal to the calculated isentropic efficiency of the cooling compressor **134** at block **297**, the controller **180** may send instructions to operate the cooling compressor **134** and to deactivate the heating compressor **136** (and/or retain the heating compressor **136** in an inactive state), as indicated by block **266**. Conversely, in response to a determination that the calculated isentropic efficiency of the heating compressor **136** is greater than the calculated isentropic efficiency of the cooling compressor **134** at block **297**, the controller **180** may send instructions to operate the heating compressor **136** and to deactivate the cooling compressor **134** (and/or retain the cooling compressor **134** in an inactive state), as indicated by block **246**. Additionally or alternatively, the controller **180** may implement the process **296** to operate more than two compressors **132**, to operate one or more multi-stage compressors **132**, and/or to operate one or more variable speed compressors **132** in accordance with the aforementioned techniques.

(83) FIG. **14** is a schematic of an embodiment of the HVAC system **100**, illustrating the heat pump **102** having a split configuration **300**. In the split configuration **300**, the heat pump **102** may include

an outdoor unit **302** having the compressor system **130**, the reversing valve **150**, the expansion device **120**, the second fan **118**, and/or the second heat exchanger **106**, for example. Moreover, in the split configuration **300**, the heat pump **102** may include an indoor unit **304** (e.g., positioned within a building or conditioned space) having an additional expansion valve **301**, the first heat exchanger **104**, and the first fan **116**, for example. Thus, the outdoor unit **302** and the indoor unit **304** may include portions of the HVAC system **100** that are disposed at different locations with respect to one another. In particular, the outdoor unit **302** may be positioned in the ambient environment **112**, while the indoor unit **304** may be positioned within the thermal load **110** and/or adjacent to the thermal load **110** (e.g., a room, an attic, a basement, or area adjacent to the space conditioned by the HVAC system **100**).

(84) A portion of the refrigerant circuit **108** included in the outdoor unit **302** may be fluidly coupled to a remaining portion of the refrigerant circuit **108** included in the indoor unit **304** via connection portions **310** (e.g., conduits) of the refrigerant circuit **108**. In the illustrated embodiment, the reversing valve **150** is positioned in the first configuration **164** to enable operation of the heat pump **102** in the cooling mode. FIG. **15** is a schematic of an embodiment of the heat pump **102** having the split configuration **300**, illustrating the reversing valve **150** in the second configuration **170**, thereby enabling operation of the heat pump **102** in the heating mode. The embodiments illustrated in FIGS. **14** and **15** may be configured to operate utilizing the techniques discussed in detail above.

(85) FIG. **16** is a schematic of an embodiment of a portion of the HVAC system **100** that includes a dual heat pump system **350** (e.g., a multi-system heat pump, a multi-circuit heat pump). The dual heat pump system **350** includes the heat pump **102** and an additional heat pump **352**. The heat pump **102** includes an embodiment of the compressor system **130** (e.g., a first compressor system) having the cooling compressor **134** and an additional cooling compressor **354** that are fluidly coupled to the refrigerant circuit **108** (e.g., relative to a flow of refrigerant through the compressors **134**, **354**) in a parallel configuration. The cooling compressor **134** and the additional cooling compressor **354** may include operational characteristics (e.g., volume ratio, volume index, volume geometry, built-in volume ratio, etc.) that are tailored (e.g., selected) to enhance operation (e.g., efficiency) of the heat pump **102** in the cooling mode. The additional heat pump **352** includes an additional refrigerant circuit **356** having an additional expansion device **358**, a first additional heat exchanger **360**, a second additional heat exchanger **362**, an additional fan **364**, an additional reversing valve **366**, and an additional compressor system **368** (e.g., a second compressor system) that includes the heating compressor **136** and an additional heating compressor **370** arranged in a parallel configuration (e.g., relative to a flow of refrigerant through the compressors **136**, **370**). The heating compressor **136** and the additional heating compressor **370** may include operational characteristics (e.g., volume ratio, volume index, volume geometry, etc.) that are tailored (e.g., selected) to enhance operation of the additional heat pump **352** in a heating mode. The first fan **116** may be configured to draw a fluid flow (e.g., an air flow) across the first heat exchanger **104**, the first additional heat exchanger **360**, or both, to facilitate thermal regulation of the thermal load **110**.

(86) In a cooling mode of the dual heat pump system **350**, the heat pump **102** may be operated to cool the thermal load **110** in accordance with the aforementioned techniques, while the additional heat pump **352** (e.g., the additional compressor system **368**) may be idle (e.g., inactive). That is, in the cooling mode of the dual heat pump system **350**, the reversing valve **150** may be in the first configuration **164** and the cooling compressor **134**, the additional cooling compressor **354**, or both, may be operational to facilitate cooling of the thermal load **110** via the first heat exchanger **104**.

(87) In a first heating mode of the dual heat pump system **350** (e.g., operation of the HVAC system **100** in response to a mild or moderate demand for heating), the heat pump **102** (e.g., the compressor system **130**) may be idle (e.g., inactive) and the additional heat pump **352** may be active (e.g., operating) to heat the thermal load **110** using the first additional heat exchanger **360**. Particularly, in the moderate heating mode, the heating compressor **136**, the additional heating compressor **370**, or both, may be operational to facilitate heating of the thermal load **110** via the

first additional heat exchanger **360**.

(88) In a second heating mode of the dual heat pump system **350** (e.g., operation of the HVAC system **100** in response to a moderate or high demand for heating), both the heat pump **102** and the additional heat pump **352** may be active to facilitate heating of the thermal load **110** via the first heat exchanger **104** and the first additional heat exchanger **360**. As such, it should be understood that, in the high heating mode of the dual heat pump system **350**, the reversing valve **150** is in the second configuration **170** and the first cooling compressor **134**, the additional cooling compressor **354**, or both, may be operational to facilitate heating of the thermal load **110** via the first heat exchanger **104**, in addition to the heating provided via the first additional heat exchanger **360** of the additional heat pump **352**. A portion of or all of the components of the dual heat pump system **350** may be communicatively coupled to the controller **180** to enable the controller **180** to operate (e.g., control) various components of the dual heat pump system **350** in accordance with the presently disclosed techniques. It should be understood that, although the dual heat pump system **350** includes two heat pumps (e.g., the heat pump **102**, the additional heat pump **352** in the illustrated embodiment, in other embodiments, the dual heat pump system **350** may include any suitable quantity of individual heat pumps. For example, the dual heat pump system **350** may include a multi-circuit heat pump system having two, three, four, five, six, or more than six separate refrigerant circuits (e.g., heat pumps) that may each be configured for improved operation in cooling modes and heating modes in accordance with the present techniques

(89) As set forth above, embodiments of the present disclosure may provide one or more technical effects useful for adjusting operation of a compressor or multiple compressors of a heat pump system based on one or more operational parameters of the heat pump system to adjust compression and/or circulation of refrigerant along a refrigerant circuit of the heat pump in a manner that enhances an overall operational efficiency of the heat pump. It should be understood that the technical effects and technical problems in the specification are examples and are not limiting. Indeed, it should be noted that the embodiments described in the specification may have other technical effects and can solve other technical problems.

(90) While only certain features and embodiments have been illustrated and described, many modifications and changes may occur to those skilled in the art, such as variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, such as temperatures and pressures, mounting arrangements, use of materials, colors, orientations, and so forth, without materially departing from the novel teachings and advantages of the subject matter recited in the claims. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the disclosure.

(91) Furthermore, in an effort to provide a concise description of the exemplary embodiments, all features of an actual implementation may not have been described, such as those unrelated to the presently contemplated best mode, or those unrelated to enablement. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation specific decisions may be made. Such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure, without undue experimentation.

(92) The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . .” or “step for [perform]ing [a function] . . .”, it is intended that such elements are to

be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

Claims

1. A heat pump for a heating, ventilation, and air conditioning (HVAC) system, comprising: a compressor system configured to direct a refrigerant flow along a refrigerant circuit of the heat pump, wherein the compressor system comprises a first compressor and a second compressor arranged in parallel with one another relative to a direction of the refrigerant flow through the compressor system, the first compressor comprises a first volume index, and the second compressor comprises a second volume index different from the first volume index; and a controller communicatively coupled to the first compressor and the second compressor and configured to selectively operate the first compressor, the second compressor, or both based on an operating mode of the heat pump, wherein the controller is configured to operate the first compressor and the second compressor in response to a determination that a temperature differential between a temperature within a space conditioned by the heat pump and a set point temperature of the space is greater than a threshold amount.
2. The heat pump of claim 1, wherein the heat pump is a reverse-cycle heat pump.
3. The heat pump of claim 1, wherein the second volume index is greater than the first volume index.
4. The heat pump of claim 3, wherein, in a cooling operating mode of the heat pump, the controller is configured to operate the first compressor and to suspend operation of the second compressor.
5. The heat pump of claim 3, wherein, in a heating operating mode of the heat pump, the controller is configured to operate the second compressor and to suspend operation of the first compressor.
6. The heat pump of claim 3, wherein, in response to a determination that the temperature differential between the temperature within the space conditioned by the heat pump and the set point temperature of the space is less than the threshold amount, the controller is configured to: receive data indicative of an ambient temperature; compare the ambient temperature to an ambient temperature set point; and operate the first compressor and suspend operation of the second compressor in response to a determination that the ambient temperature is greater than the ambient temperature set point.
7. The heat pump of claim 6, wherein the controller is configured to operate the second compressor and suspend operation of the first compressor in response to a determination that the ambient temperature is equal to or less than the ambient temperature set point.
8. The heat pump of claim 3, wherein, in response to a determination that the temperature differential between the temperature within the space conditioned by the heat pump and the set point temperature of the space is less than the threshold amount, the controller is configured to: receive data indicative of an outdoor coil temperature; compare the outdoor coil temperature to an outdoor coil temperature set point; and operate the first compressor and suspend operation of the second compressor in response to a determination that the outdoor coil temperature is greater than the outdoor coil temperature set point.
9. The heat pump of claim 8, wherein the controller is configured to operate the second compressor and suspend operation of the first compressor in response to a determination that the outdoor coil temperature is equal to or less than the outdoor coil temperature set point.
10. The heat pump of claim 1, wherein the first compressor or the second compressor is a fixed speed compressor, a multi-stage compressor, or a variable speed compressor.
11. A heat pump for a heating, ventilation, and air conditioning (HVAC) system, comprising: a first compressor configured to direct refrigerant flow along a refrigerant circuit of the heat pump, wherein the first compressor comprises a first built-in volume ratio; a second compressor configured to direct refrigerant flow along the refrigerant circuit of the heat pump, wherein the

second compressor comprises a second built-in volume ratio; and a controller communicatively coupled to the first compressor and the second compressor and configured to selectively operate the first compressor, the second compressor, or both based on an operating mode of the heat pump, wherein the controller is configured to operate the first compressor and suspend operation of the second compressor in a cooling mode of the heat pump and configured to operate the second compressor and suspend operation of the first compressor in a heating mode of the heat pump, wherein the first compressor and the second compressor are arranged in parallel with one another relative to a direction of refrigerant flow through the refrigerant circuit, and the second built-in volume ratio is greater than the first built-in volume ratio.

12. The heat pump of claim 11, comprising a reversing valve, wherein the reversing valve is adjustable between a first position and a second position, the reversing valve is configured to direct refrigerant flow to an outdoor heat exchanger of the refrigerant circuit in the first position and in the cooling mode of the heat pump, and the reversing valve is configured to direct refrigerant flow to an indoor heat exchanger of the refrigerant circuit in the second position and in the heating mode of the heat pump.

13. The heat pump of claim 12, comprising the outdoor heat exchanger and the indoor heat exchanger, wherein the first compressor, the second compressor, the reversing valve, and the outdoor heat exchanger are packaged within an outdoor unit of the heat pump, and the indoor heat exchanger is packaged within an indoor unit of the heat pump.

14. The heat pump of claim 11, wherein the controller is configured to: operate the first compressor and suspend operation of the second compressor in response to a determination that a set point temperature of a conditioned space is less than a detected temperature of the conditioned space; and operate the second compressor and suspend operation of the first compressor in response to a determination that the set point temperature of the conditioned space is greater than the detected temperature of the conditioned space.

15. The heat pump of claim 14, wherein the controller is configured to: operate the second compressor and suspend operation of the first compressor in response to a determination that the set point temperature of the conditioned space is greater than the detected temperature of the conditioned space by a first temperature differential less than a threshold amount; and operate the second compressor and the first compressor in response to a determination that the set point temperature of the conditioned space is greater than the detected temperature of the conditioned space by a second temperature differential greater than the threshold amount.

16. The heat pump of claim 11, wherein the first compressor or the second compressor is a fixed speed compressor, a multi-stage compressor, or a variable speed compressor.

17. A heat pump for a heating, ventilation, and air conditioning (HVAC) system, comprising: a refrigerant circuit; a compressor system disposed along the refrigerant circuit and configured to direct a refrigerant along the refrigerant circuit, wherein the compressor system comprises a first compressor and a second compressor arranged in parallel with one another relative to a flow direction of the refrigerant through the compressor system, the first compressor comprises a first volume index, and the second compressor comprises a second volume index greater than the first volume index; a first heat exchanger disposed along the refrigerant circuit and configured to place the refrigerant in a heat exchange relationship with a supply air flow; a second heat exchanger disposed along the refrigerant circuit and configured to place the refrigerant in a heat exchange relationship with an ambient air flow; a reversing valve configured to direct the refrigerant from the compressor system to the second heat exchanger in a cooling mode of the heat pump and configured to direct the refrigerant from the compressor system to the first heat exchanger in a heating mode of the heat pump; and a controller communicatively coupled to the first compressor and the second compressor and configured to operate the first compressor and suspend operation of the second compressor in the cooling mode.

18. The heat pump of claim 17, wherein the controller is configured to operate the second

compressor and suspend operation of the first compressor in the heating mode.

19. The heat pump of claim 17, wherein the controller is configured to: operate in the heating mode in response to a determination that a temperature of a conditioned space is less than a set point temperature of the conditioned space; operate the first compressor and the second compressor in response to a determination that a temperature differential between the temperature of the conditioned space and the set point temperature is greater than or equal to a threshold amount; and operate the first compressor or the second compressor in response to a determination that the temperature differential between the temperature of the conditioned space and the set point temperature is less than the threshold amount.

20. The heat pump of claim 19, wherein the controller is configured to: operate the first compressor and suspend operation of the second compressor in response to the determination that the temperature differential between the temperature of the conditioned space and the set point temperature is less than the threshold amount and in response to a determination that an operating parameter of the heat pump is greater than or equal to a set point value; and operate the second compressor and suspend operation of the first compressor in response to the determination that the temperature differential between the temperature of the conditioned space and the set point temperature is less than the threshold amount and in response to a determination that the operating parameter of the heat pump is less than the set point value, wherein the operating parameter of the heat pump is an ambient temperature, a temperature of the second heat exchanger, a suction pressure of the refrigerant, or a suction saturation temperature of the refrigerant.
